Expected value

1 Consider the table:

x	0	1	2	3	4
$P(X = x)$	0.1	0.45	0.15	0.05	0.25

a State whether the table represents a discrete probability distribution. Explain your answer.

b State the most likely outcome for the random variable X.

c Find the expected value, $E(X)$.

2 For each of the following probability distribution tables for the random variable X:

i State the most likely outcome. **ii** Find the expected value, $E(X)$.

a

x	-1	2	5	8
$P(X = x)$	0.05	0.1	0.25	0.6

b

x	-6	-4	-2	0	2	4	6
$P(X = x)$	$\frac{2}{32}$	$\frac{4}{32}$	$\frac{6}{32}$	$\frac{8}{32}$	$\frac{6}{32}$	$\frac{4}{32}$	$\frac{2}{32}$

c

x	2	4	6	8	10	12	14	16
$P(X = x)$	0.04	0.03	0.07	0.35	0.11	0.08	0.19	0.13

3 X is a uniform discrete random variable that takes on the values 1, 2, 3 or 4.

a Construct the probability distribution table for X.

b Find $E(X)$.

4 Consider the probability density function defined by:

$$P(X = x) = \begin{cases} \frac{x}{25}; \ x = 3, 4, 5, 6, 7 \\ 0, \text{ otherwise} \end{cases}$$

a Construct the probability distribution table for X.

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b Calculate the expected value of the distribution.

5 The probability density function represents the distribution of a discrete random variable:

$$P(X = x) = \begin{cases} xk;\ x = 2,\ 3,\ 4,\ 5,\ 6 \\ 0,\ \text{otherwise} \end{cases}$$

a Find the value of k.

b Construct the probability distribution table for X.

c Calculate the expected value of the distribution.

6 Let X be a random variable with the following probability distribution table:

x	2	3	4	5	6
$P(X = x)$	0.2	$0.6p^2$	0.1	$1 - p$	0.1

a Find the possible values of p.

b For each possible value of p, find:

i $E(X)$ **ii** $P(X \geq E(X))$

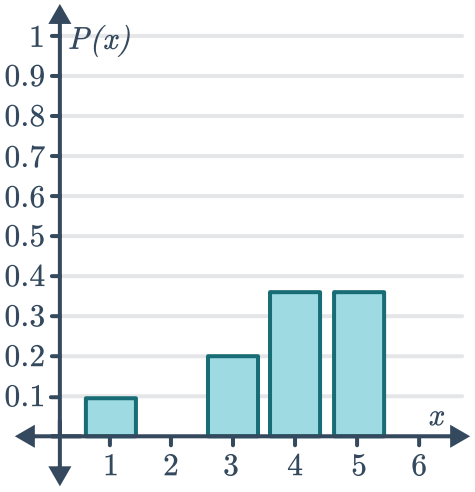
7 For each of the probability distribution graphs below:

i List the possible outcomes of X.

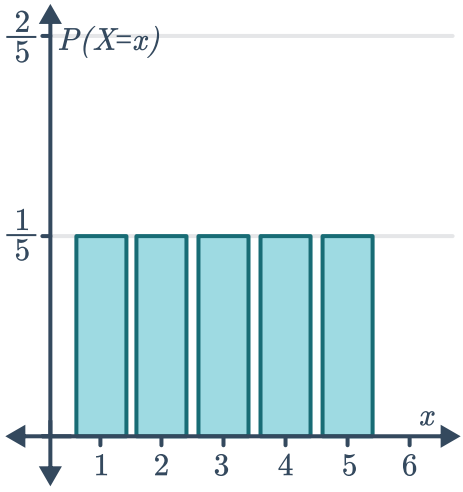
ii Classify the distribution as uniform or non-uniform.

iii Calculate the expected value of the distribution.

a



b



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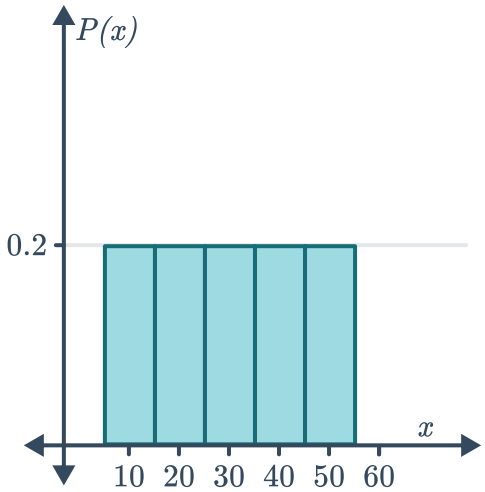
8 For each of the probability distribution graphs below:

i Calculate the expected value of the distribution.

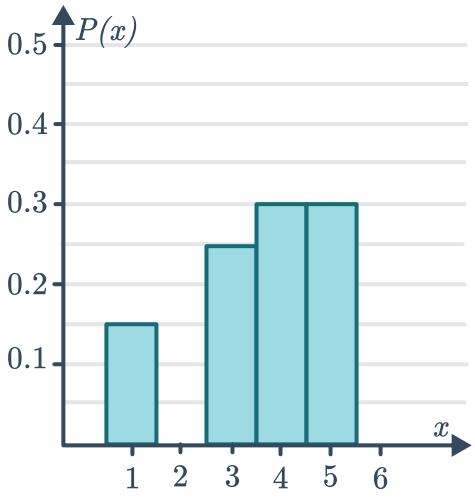
ii Determine if the median of the distribution is higher, lower or equal to the mean.

ii State whether the distribution is negatively skewed, postively skewed or symmetrical.

a



b



Variance and standard deviation

9 Consider the following probability distribution table for X:

x	0	1	2	3	4	5
$P(X=x)$	0.14	0.24	0.13	0.23	0.13	0.13

a Find $E(X)$.

b Find $\text{Var}(X)$.

10 Consider the table:

x	0	1	2	3	4
$P(X=x)$	0.12	0.15	0.22	0.23	0.28

a Explain why the table represents a discrete probability distribution.

b Find $E(X)$.

c Find the variance of X.

d Hence, find the standard deviation to two decimal places.

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11 Consider the following probability distribution table for X:

x	1	3	5	7	9
$P(X = x)$	0.1	0.08	0.2	0.07	0.55

a Find $E(X)$.

b Find $E(X^2)$.

c Find $E(2X + 5)$.

d Find $E\left(\frac{1}{X}\right)$ to two decimal places.

e Find $\text{Var}(X)$.

12 Consider the following probability distribution table for X:

x	1	3	5	7	9
$P(X = x)$	$\frac{1}{12}$	k	$\frac{1}{2}$	$\frac{1}{20}$	$\frac{1}{5}$

a Find the value of k.

b Find $E(X)$.

c Find $\text{Var}(X)$.

13 For each of the following probability distribution tables:

i Find the value of k.

ii Find the mode.

iii Find the expected value.

iv Find the variance.

v Find the standard deviation to two decimal places.

a

x	3	5	7	9	11
$P(X = x)$	0.15	0.2	k	0.1	0.3

b

x	1	3	5	7	9
$P(X = x)$	$5k$	k	$6k$	$2k$	$3k$

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14 The table below represents a discrete probability distribution:

x	1	2	3	4	5
$P(X=x)$	0.05	m	0.1	n	0.15

a Use a property of probability distributions to express m in terms of n.

b Use the fact that $E(X) = 3.1$ to express m in terms of n.

c Hence, solve for n.

d Hence, solve for m.

e Calculate the standard deviation to one decimal place.

15 For each of the probability density functions described below:

i Construct the probability distribution table for X.

ii Find the expected value of the distribution.

iii Find the standard deviation of the distribution. Round your answer to two decimal places where necessary.

a $P(X=x) = \begin{cases} \frac{x}{10}; x = 1, 2, 3, 4 \\ 0, \text{ otherwise} \end{cases}$

b $P(X=x) = \begin{cases} \frac{7-x}{19}; x = 0, 1, 3, 5 \\ 0, \text{ otherwise} \end{cases}$

c $P(X=x) = \begin{cases} \frac{x^2-1}{50}; x = 2, 3, 4, 5 \\ 0, \text{ otherwise} \end{cases}$

16 For each of the probability density functions described below:

i Find the value of k.

ii Construct the probability distribution table for X.

iii Find the expected value of the distribution.

iv Find the standard deviation of the distribution. Round your answer to one decimal place.

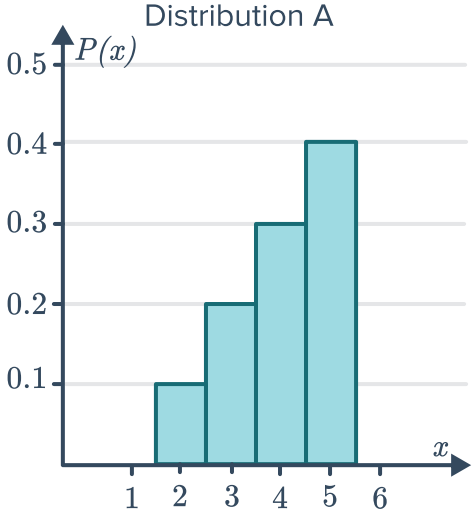
a $P(X=x) = \begin{cases} \frac{x^2}{k}; x = 1, 2, 5, 7 \\ 0, \text{ otherwise} \end{cases}$

b $P(X=x) = \begin{cases} k(6-x); x = 0, 1, 2, 3, 4 \\ 0, \text{ otherwise} \end{cases}$

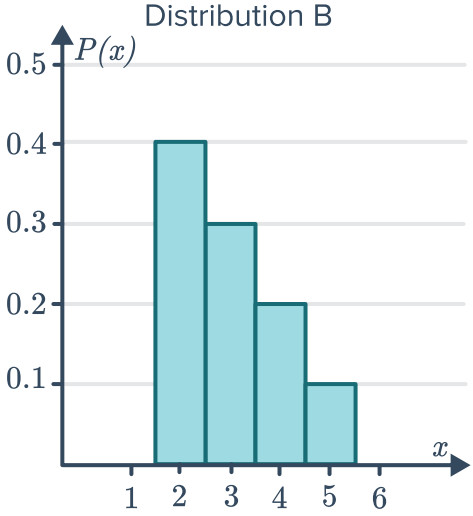
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17 Consider the discrete probability distribution graphs below:

Distribution A



Distribution B



a Calculate the expected value for each distribution.

b Describe the shape of each distribution.

c For each distribution determine if the median of the distribution is higher, lower or equal to the mean.

d Calculate the variance of distribution A.

e Hence, calculate the standard deviation of distribution A.

f Would you expect the standard deviation of distribution B to be greater than, less than or equal to the standard deviation of distribution A? Explain your answer.

Applications

18 The Vaucluse Vigilantes and the Woollahra Weasels play a series of basketball matches and the first to win four games wins the series. The following distribution shows the probabilities of the total number of games that will be played in the series:

X	4	5	6	7
$p(X = x)$	0.11	0.34	0.33	0.22

Find the expected number of games that will be played in the series.

19 A game uses a spinner with numbers from 1 to 12, each with equal outcome. A player wins $\$6$ if the outcome is greater than 9, $\$4$ if the outcome is less than 5 and loses $\$2$ for any other outcome.

a Let X be the return from one game. Construct the probability distribution table for X.

b Find the expected return in dollars.

c Calculate the exact standard deviation.

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20 A deck of cards has cards numbered 8 through 16. A player draws a card at random. A player wins $\$3$ if the card is odd and loses $\$3$ if it's even.
Let X be the winnings of the player. Find the expected return in dollars.

21 Xavier tosses two coins. He wins $\$10$ for two tails, $\$5$ for two heads, and nothing for a head and a tail.

a Find the expected value of this game.

b If the game costs $\$5$ to play, will Xavier likely to win or lose in the long run? Explain your answer.

22 A fair die is rolled. If it lands on an odd number, then the score is the number. But if it lands on an even number, then the score is 0. Let X be the score when the die is rolled once.

a Find $E(X)$.

b Find $\text{Var}(X)$.

23 A cat has a litter of three kittens. Each kitten is equally likely to be born male or female.

a Construct a tree diagram to represent all possible combinations of the sex of these three kittens.

b Let M be the number of male kittens born to this cat. Construct the probability distribution table for M.

c State the expected number of male kittens born.

24 Xanthe is sitting a multiple choice quiz consisting of 3 questions, each with 4 possible answers. Xanthe hasn't studied for the quiz so she will guess the answer to each question at random.

a Construct a probability tree diagram to represent all possible combinations of which questions Xanthe got right and wrong.

b Let X be the number of correctly answered questions. Construct a probability distribution table for X.

c Determine the most likely number of questions Xanthe guessed correctly.

d Calculate how many questions Xanthe was expected to answer correctly by guessing.

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25 Let the set of numbers $n = 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12$ be a sample space, and let L be a random variable over the sample space defined as:

$$L(n) = \text{number of letters in the English word for the number } n$$

a Each outcome of the sample space is equally likely to occur. Complete the probability distribution table for L:

l	3	4	5	6
$P(L = l)$				

b Calculate the average number of letters in the English words for numbers 1 to 12.

c Find the exact standard deviation of letters in the English words for numbers 1 to 12.

26 When calling your internet service provider, the probability of being placed in a queue with x customers ahead of you is given by:

$$p(x) = k(0.24)^x ; x = 0, 1, 2, 3, \ldots$$

where k is a positive constant.

a Given that the probability of there being 2 people in the queue before you is 0.043776, solve for k.

b This distribution is a geometric distribution and its expected value can be calculated by:

$$E(X) = \frac{\text{Probability of there being a customer before you in the queue}}{1 - \text{Probability of there being a customer before you in the queue}}$$

Calculate the expected number of customers in the queue before you are served. Give your answer to two decimal places.

27 In a large batch of microphones, 10% are known to be defective. An online retailer mails out 3 microphones, randomly selected from the batch.

a Construct a probability tree diagram to represent all possible combinations of defective, D, and working, W, microphones sent out.

b Let X be the number of defective microphones sent out. Construct the probability distribution table for X.

c Find the expected value of X.

d The online retailer gives gift vouchers to customers who return their defective microphones. The value of gift voucher in dollars is given by $C(X) = 20X$. Calculate the expected value of the gift voucher for this batch.

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28 To open a tutoring business, Quentin does some market research into the profit or loss of similar businesses in his area in the first year. His findings are summarised in the table shown:

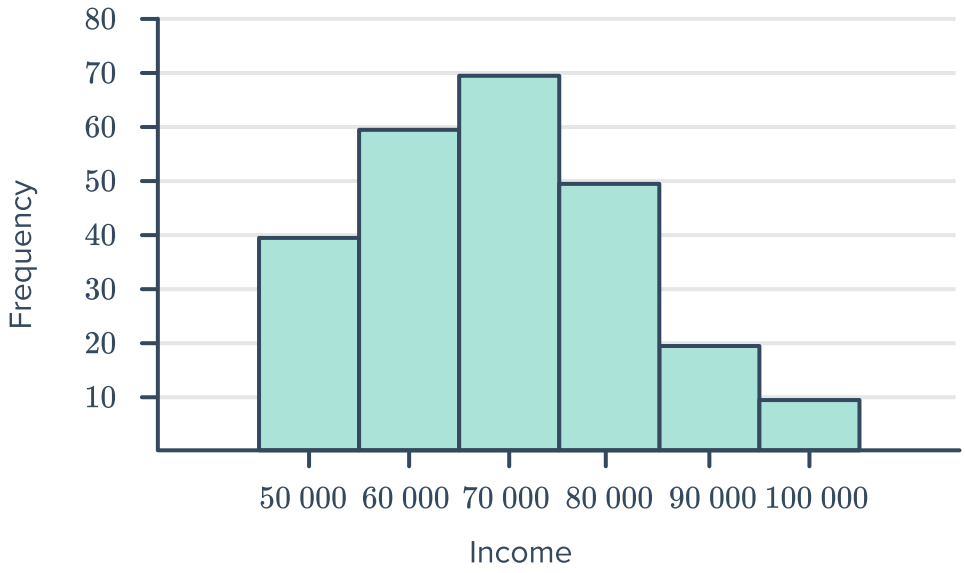
a Does this table represent a discrete probability distribution? Explain your answer.

b Calculate Quentin's expected profit.

c Calculate the variance in his expected profit.

Earnings	Probability
$-\$1000$	0.15
$\$0$	0.1
$\$3000$	0.5
$\$5000$	0.25

29 A school is surveying its past students on their income, 15 years after they've left school. The earnings, rounded to the nearest $\$10\,000$, for the leavers of 2001 are shown in the graph. The school believes these results are fairly indicative of the earnings of all their students and will use this data to make further predictions.
Let I represent the earnings of past students from this school.



a Complete the probability distribution table below:

i	$\$50\,000$	$\$60\,000$	$\$70\,000$	$\$80\,000$	$\$90\,000$	$\$100\,000$
$P(I = i)$						

b Calculate the expected earnings of a student from this school.

c Calculate the variance in student income, correct to two decimal places.

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